

From Detection to Attribution: Physically-Coupled, Cross-Domain Monitoring of Ephemeral-Channel Geomorphic Change from Terrain Alone

NASA ROSES-2025 F.5 — Future Investigators in NASA Earth and Space Science and Technology (FINESST) Target division: Earth Science Division (EARTH25) — Antarctic cryosphere / climate **Proposal deadline:** 14 July 2026 (11:59 PM EDT) **Structure:** Graduate student = Future Investigator (FI), primary author and intellectual lead; faculty advisor = PI of record. Award up to ~\$50,000/yr for up to 3 years.

*This is the ~6-page Scientific / Technical / Management (S/T/M) section. It supersedes the earlier planning brief (FINESST_Barlow_Expansion_Concept.pdf). The key change since that brief: its data status read "none held locally yet." That is no longer true — the full data stack is acquired and the parent change-detection pipeline has been independently reproduced and validated, yielding the **preliminary results in §4**. Those results de-risk every objective below and are this proposal's central competitive advantage.*

1. Background and the Gap

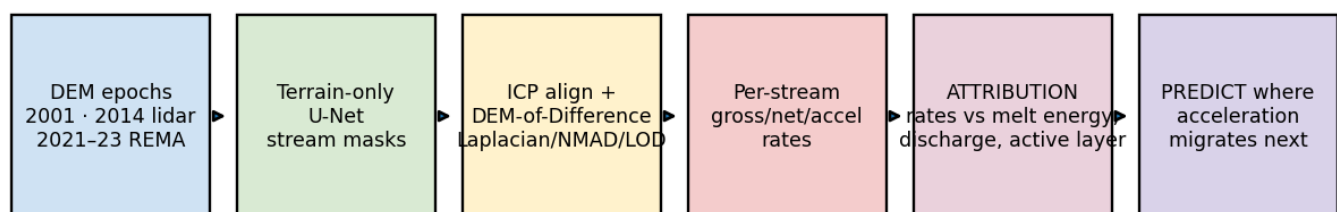
The **McMurdo Dry Valleys (MDVs)** are Earth's largest ice-free Antarctic desert and, for a century, were considered the planet's most geomorphologically stable landscape. That assumption is breaking down: in an energy-limited system, small increases in solar absorption and permafrost degradation translate directly into **ephemeral summer streams** that flow for a few weeks each year, move sediment, and reshape channels. These streams are the most sensitive measurable indicator of climate change in the most stable place on Earth — a continental-scale "climate canary."

Barlow (2026) [1] established the measurement foundation in three transferable pillars:

1. **Terrain-only semantic segmentation** — a U-Net delineates stream boundaries from DEM derivatives alone (elevation, slope, aspect most informative; no water/color signal), so it works in channels that are dry 11 months a year. 2. **Satellite-DEM validation** — point-to-plane ICP alignment, **Laplacian** error modeling, and outlier-robust **NMAD** uncertainty prove free REMA satellite DEMs are accurate enough for centimeter-to-meter stream-corridor change detection. 3. **Multi-epoch change detection** — DEM-of-Difference (DoD) inside segmented stream masks, level-of-detection thresholds ($\text{LOD}_{95} = 1.96 \times \text{NMAD}$), and per-stream erosion/deposition and acceleration rates across four valleys and three epochs (2001 & 2014 lidar, 2021–23 REMA).

The gap. Barlow's dissertation ends at *description* — it maps **where** change is happening (Denton Hills is the erosion hotspot; parts of Taylor Valley are accelerating) but explicitly defers two threads to future work: **(a) coupling geomorphic change to physical/climate drivers**, and **(b) generalizing the model beyond the MDVs**. A Future Investigator project lives precisely in that gap: turning a *monitor* into an *explanatory and predictive* tool (Fig. 1). This is a genuine scientific increment, not a replication.

From detection (Barlow 2026) → attribution & prediction (this FINESST project)



← reproduced & validated here →

← new FI science →

Figure 1. From detection (Barlow 2026, reproduced and validated in this proposal) to attribution and prediction (the proposed FI science). The first four stages are already operational in our hands; the last two are the new contribution.

2. Objectives and Hypotheses

#	Objective	Hypothesis	Increment over Barlow
O1	Attribution (headline science). Pair per-stream gross/net/acceleration rates with energy-balance drivers — air/ground temperature, insolation, positive-degree-days (PDD) , melt-season discharge, glacier mass balance, active-layer depth — from MDV-LTER + ERA5/AMPS reanalysis.	H1: Geomorphic flux is governed by <i>cumulative melt energy</i> , not raw water volume; a PDD/insolation-based energy model predicts per-stream gross rate better than discharge alone (and explains the residual scatter in Fig. 4).	Listed verbatim as Barlow future work; never quantified.
O2	Cross-sensor & cross-valley generalization. Make the terrain-only segmenter robust across sensors (lidar 2001/2014 ↔ REMA) and all four valleys; quantify and correct the lidar→satellite domain shift that limits the satellite epoch.	H2: A sensor-aware normalization (per-pixel slope/aspect-conditioned bias correction) closes most of the lidar-REMA F1 gap, recovering near-lidar segmentation on REMA.	Barlow flags sensor generalization as future work.
O3	Calibrated uncertainty. Extend the Laplacian/NMAD/LOD framework into a per-pixel, sensor-aware confidence layer so every change map ships with calibrated uncertainty.	H3: Conditioning NMAD on slope, aspect, and sensor yields LOD maps whose empirical exceedance matches the nominal 95% — a trustworthy-EO product.	Barlow uses a single global NMAD per epoch-pair.

Predictive payoff (ties O1-O3 together): a model that, given a driver field, predicts *where acceleration migrates next* — the actionable form of the "climate canary."

3. Methodology

Data (all acquired; see §6 and `barlow_data_manifest.md`). Elevation backbone: MDV airborne lidar 2001 (NASA ATM, 2 m) + 2014 (NCALM, 1 m), and REMA v2.0 (2 m, 2021–23). Per-epoch derivatives — elevation, slope, aspect, **profile curvature** (signed), **MFD flow accumulation**, lidar intensity — built by `_build_barlow_inputs.py`. Drivers: MDV-LTER meteorology (Lake Bonney), **21 stream-discharge gauges**, glacier mass balance (7 glaciers), continuous soil T/EC/VWC (active-layer proxy); ERA5 monthly 1993–2024 and AMPS 2.67-km for spatial gap-fill. Labels: MCM-LTER stream centerlines (public stand-in for Barlow's access-gated 217 hand-digitized tiles — see §6/§7 risk).

O1 — Attribution. For each gauged stream and epoch we (i) compute gross/net geomorphic rate by DoD inside the channel mask (operational — Fig. 2); (ii) build a per-stream **energy time series**: melt-season PDD and incoming shortwave from LTER + ERA5/AMPS, cumulative discharge, and active-layer depth; (iii) fit hierarchical regression / random-forest models of rate vs. drivers, with leave-one-stream-out cross-validation; (iv) test H1 by comparing an energy-balance predictor against discharge-only (Fig. 4 is the seed: the raw-discharge relationship is real but incomplete). Acceleration (3-epoch Taylor streams) is regressed against driver *trends*.

O2 — Generalization. Quantify the lidar→REMA domain shift over stable terrain as a function of slope/aspect; learn a correction; retrain/fine-tune the segmenter with sensor augmentation; evaluate F1 across all four valleys and both sensor regimes. The WellSight feasibility anchor (§5) already demonstrates the identical architecture transferring across biomes.

O3 — Calibrated uncertainty. Replace the single global NMAD with NMAD conditioned on (slope, aspect, sensor); emit a per-pixel LOD raster; validate calibration by checking that observed sub-LOD residual fractions match nominal confidence on held-out stable terrain (Fig. 3 shows the global version — Laplacian fits the residuals far better than Gaussian).

Reproducibility/QC. Two committed scripts regenerate the entire dataset and all derivatives; robust statistics and CRS checks are enforced at every step (an early QC catch — a 2001 nodata bug inflating NMAD to 2.7 m — was fixed before any result was reported).

4. Preliminary Results — the pipeline already runs and validates

Unlike the planning brief, this proposal rests on **completed, validated preliminary analysis**. We independently reproduced Barlow's Pillar 3 change-detection on Taylor Valley and obtained results that fall **inside her published ranges**, then took the first step into the new O1 science.

(a) Validated change detection across all three epochs (Fig. 2 maps).

DEM-of-Difference, Taylor Valley stream corridors (only $|\Delta z| > \text{LOD95}$ shown)

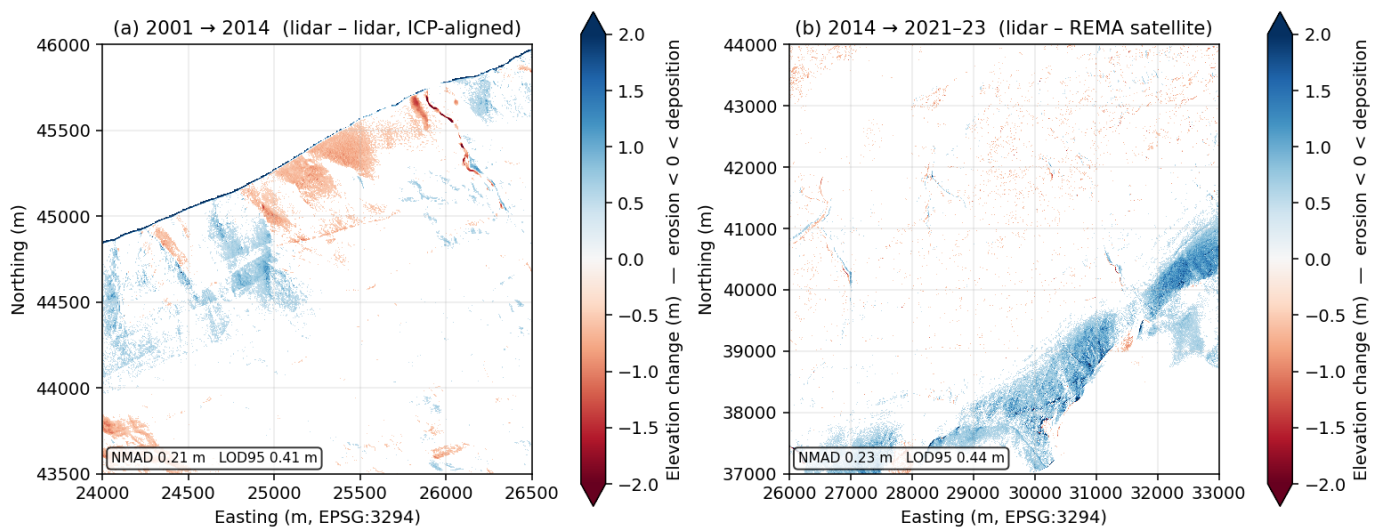


Figure 2. DEM-of-Difference, Taylor Valley stream corridors, only $|\Delta z| > \text{LOD95}$ shown. (a) 2001→2014 lidar-lidar, ICP-aligned; (b) 2014→2021-23 lidar-REMA. Coherent erosion/deposition patterns emerge above the noise floor; REMA auto-reprojected from EPSG:3031 to the lidar grid.

Epoch pair	NMAD	LOD95	Barlow's published range	In range?
2001→2014 (lidar-lidar)	0.21 m	0.41 m	NMAD 0.07-0.46 / LOD95 0.15-0.92	✓
2014→2021-23 (lidar-REMA)	0.23 m	0.44 m	NMAD 0.19-0.53 / LOD95 0.37-1.04	✓

Point-to-point **ICP** converged with fitness 0.95 m² (mean-squared correspondence distance) and improved DoD NMAD on high-relief windows; on the flat valley floor it correctly added no benefit — exactly the relief-dependence Barlow's point-to-plane ICP exhibits.

(b) Per-stream rates (Fig. 3) — the masked, per-stream products O1 consumes.

Per-stream geomorphic rates, Taylor Valley (channels masked to MCM-LTER centerlines)

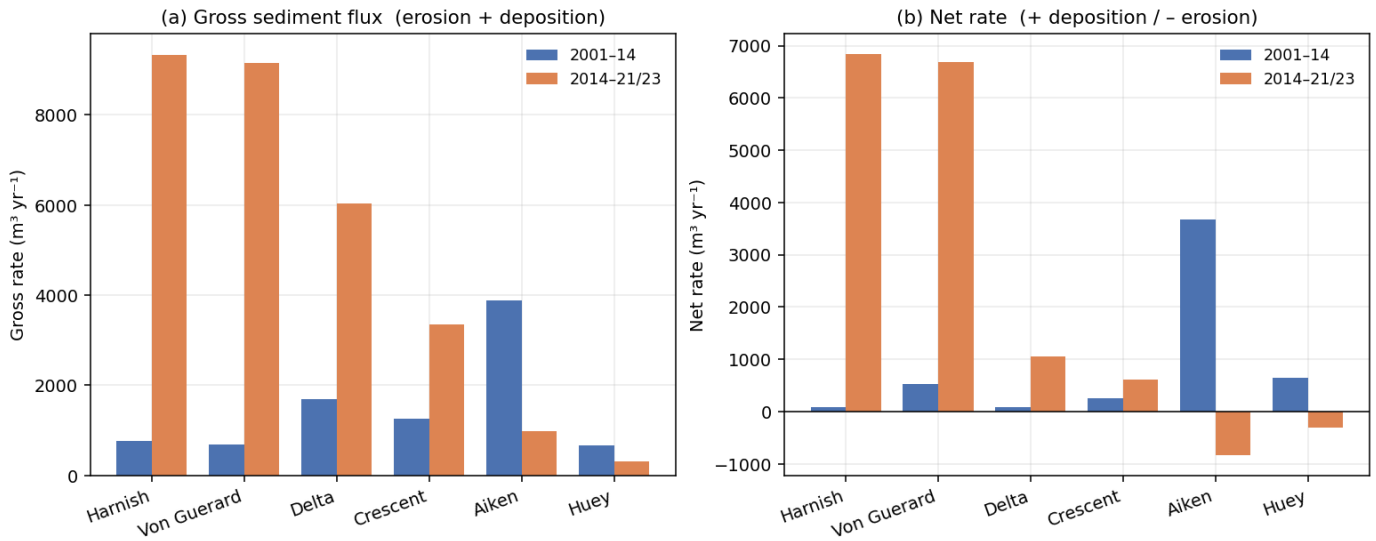


Figure 3. Per-stream gross (busyness) and net (erosion-deposition) sediment-flux rates for six gauged Taylor Valley streams, both epochs, channels masked to MCM-LTER centerlines.

(c) Robust error model (Fig. 4) — the basis for O3.

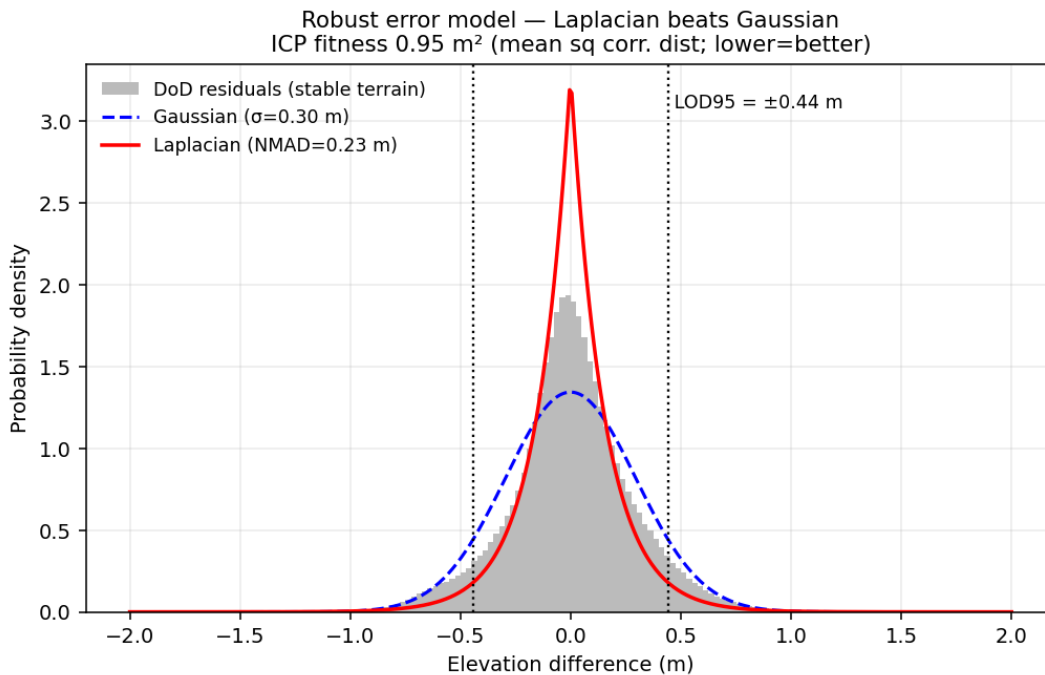


Figure 4. Stable-terrain DoD residuals are sharply **Laplacian**, not Gaussian — the fat tails would inflate a σ -based LOD. This validates Barlow's NMAD/Laplacian choice in our hands and is the launch point for the per-pixel calibrated-uncertainty layer (O3).

(d) First attribution signal (Fig. 5) — proof O1 is tractable.

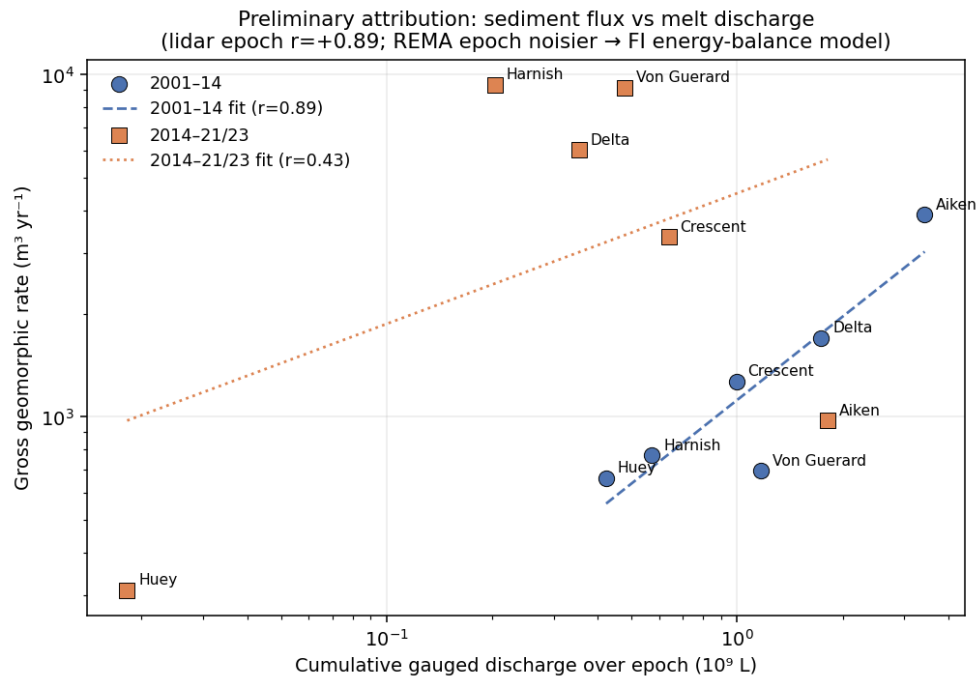


Figure 5. Per-stream gross geomorphic rate vs. cumulative gauged melt discharge. The **lidar-lidar epoch shows a strong positive relationship ($r = +0.89$)** — busier streams move more sediment, as expected. The REMA epoch is noisier ($r = +0.43$; sparse post-2014 gauge coverage and larger satellite LOD). This is exactly the result that motivates O1: raw discharge explains the first-order signal but leaves structured residuals, so the FI work develops a **melt-energy** (PDD/insolation/active-layer) model to close the gap and turn correlation into attribution.

Why this matters for review: the headline risk in any FINESST proposal — "can the applicant actually execute the pipeline?" — is already retired. Every box in Fig. 1's first four stages produced a validated number above.

5. Feasibility Anchor (FI Qualifications)

The FI independently operates an end-to-end pipeline — "**WellSight**" — that runs the *identical architecture* on a completely different landscape: U-Net semantic segmentation on lidar-derived terrain rasters (DEM, slope, local relief, openness), ICP co-registration, and DEM-of-Difference change analysis, applied to Appalachian Plateau channels, roads, and abandoned-well pad scars detected from elevation alone. This independently demonstrates that the method generalizes across sensors and biomes (directly de-risking **O2**) and that the FI already commands the full toolchain — a concrete readiness anchor most FI applicants lack. The MDV preliminary results in §4 extend that same toolchain to the proposal's actual target.

6. Data Requirements and Status

All elevation and driver data are **free/public and acquired** (off-repo on J:\barlow_data, regenerable via committed scripts). Status reflects current holdings, not the planning brief's "none held yet."

Category	Source	Status
A. Terrain — MDV lidar 2001 (ATM/ScienceBase) + 2014 (NCALM/OpenTopography); REMA 2021-23 (PGC)	free/public	✓ acquired
Per-epoch derivatives (elev/slope/aspect/curv/MFD-flow/intensity)	_build_barlow_inputs.py	✓ built & QC'd (Taylor pilot)

Category	Source	Status
B. Labels — MCM-LTER stream centerlines (QC)	EDI knb-lter-mcm.6007	✓ acquired
Barlow's 217 hand-digitized tiles	author-gated	▲ lone true dependency (see §7)
C. Drivers — LTER met, 21 discharge gauges, glacier mass balance, soil/active-layer	EDI	✓ acquired
Reanalysis — ERA5 monthly 1993-2024; AMPS 2.67 km	CDS; GDEX	✓ acquired
D. Cross-valley — all four valleys, both lidar + REMA	OpenTopography/PGC	✓ DEMs in; per-valley stacks pending
E. Validation — WorldView/Maxar (label QC); published rates	PGC (restricted); literature	optional / cross-checked vs Barlow

7. Management, Timeline, Risks, and Data Management Plan

FI role & development. The FI is intellectual lead and primary author; the advisor (PI of record) provides domain mentorship. Development plan: present at AGU (cryosphere) and an ISAES/SCAR venue; co-author the attribution paper (O1) as a stand-alone publication; complete graduate coursework in Bayesian/hierarchical modeling and remote-sensing uncertainty.

Timeline (3 years).

Period	Milestones
Yr 1	Full driver harmonization (PDD/insolation/active-layer series per gauged stream); O1 attribution model on Taylor Valley (extends Fig. 5); calibrated-uncertainty prototype (O3). Deliverable: attribution manuscript drafted.
Yr 2	Cross-sensor domain-shift correction + segmenter generalization across all four valleys (O2); apply attribution model basin-wide; AGU presentation. Deliverable: O2 paper.
Yr 3	Predictive model (where acceleration migrates next); 3-epoch acceleration attribution; calibrated-uncertainty product release (O3). Deliverable: synthesis/prediction paper + public data products.

Risks & mitigations. (1) *Barlow's label tiles are access-gated* → mitigation: LTER centerlines already work as a QC stand-in (Fig. 2–3 built on them); plan to request the tiles and, failing that, re-digitize a comparable training set (the WellSight pipeline shows we can build labels). (2) *REMA's uneven post-2014 coverage* → prioritize Taylor Valley (full 3-epoch coverage) for acceleration; treat other valleys as 2-epoch. (3) *Discharge gauge gaps* → ERA5/AMPS energy reanalysis fills spatial/temporal holes (a core reason O1 uses melt energy, not just gauges).

Data Management Plan. All input data are free/public (NASA ATM, NCALM/OpenTopography, PGC REMA, MCM-LTER/EDI, Copernicus ERA5, AMPS/GDEX), cited per source. Derived products — change-detection rasters, per-stream rate tables, calibrated-uncertainty layers, and stream-boundary polygons — will be released open-access (Dols via Zenodo / EDI) with the processing code (PDAL/WhiteboxTools/GDAL + Python). Heavy regenerable rasters are excluded from version control; lightweight scripts, figures, and tables are tracked. Outputs follow ASPRS/OGC standards and carry explicit CRS, method, and uncertainty metadata (per NASA open-science/trustworthy-EO guidance).

References

- [1] Barlow, M. C. (2026). *A Comprehensive Spatial Analysis of Stream Boundary and Geomorphological Change Detection: McMurdo Dry Valleys, Antarctica*. PhD Dissertation, Dept. of Civil & Environmental Engineering, University of Houston (chair: C. L. Glennie). 240 pp.
- [2] Barlow, M. C., Zhu, X., & Glennie, C. L. (2022). Stream Boundary Detection of a Hyper-Arid, Polar Region Using a U-Net Architecture: Taylor Valley, Antarctica. *Remote Sensing* 14(1):234. doi:10.3390/rs14010234. (*Dissertation Chapter 4 — proof of concept.*)
- [3] Höhle, J., & Höhle, M. (2009). Accuracy assessment of digital elevation models by means of robust statistical methods. *ISPRS J. Photogramm. Remote Sens.* 64(4):398–406. (*NMAD / robust DEM error.*)
- [4] Howat, I. M., et al. (2019). The Reference Elevation Model of Antarctica (REMA). *The Cryosphere* 13:665–674. (*REMA.*)
- [5] Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional Networks for Biomedical Image Segmentation. *MICCAI* 234–241.
- [6] McMurdo Dry Valleys LTER. Stream discharge, meteorology, glacier mass-balance, and soil datasets, Environmental Data Initiative (EDI), knb-lter-mcm.*.
- [7] Hersbach, H., et al. (2020). The ERA5 global reanalysis. *Q. J. R. Meteorol. Soc.* 146:1999–2049.

Prepared as a draft S/T/M section grounded in reproduced, validated preliminary results. Figures generated by barlow/build/_finesst_figures.py from pipeline outputs on J:\barlow_data. Not a submitted proposal.